

Relativistic MOND Is a Cuscuton Theory: A Candidate de Sitter–MOND Completion, Its Structural Spine, Falsifiable Predictions, and Open Costs

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Abstract

We report the status of a candidate relativistic completion of Milgrom’s modified Newtonian dynamics (MOND) in which the acceleration scale is locked to the de Sitter/cosmological constant, $a_0 = c^2/2\pi L_{\text{dS}}$, and the modified-inertia sector is carried by a *clock* — a scalar φ entering only through the clock-projected derivative $Q = n^\mu \partial_\mu \varphi$, with n_μ the unit gradient of a primordial clock field. The programme’s central structural result is a classification theorem: **a relativistic MOND scalar carrying a canonical kinetic term cannot close the Hamiltonian (hypersurface-deformation) constraint algebra**; matching the spatial-diffeomorphism generator forces the kinetic coefficient to $A = 1/\mu$, and the residual cubic obstruction $AA'/2 = -\mu'/(2\mu^3)$ is nonzero for every strictly monotone interpolating kernel. The unique consistent branch is the one on which the MOND scalar is a **cuscuton** (non-propagating, momentum-constrained); there, constraint closure and ghost-freedom coincide. This subsumes, as one fact, a family of otherwise-separate health results — the evasion of a deep-MOND gradient instability by keeping MOND inside the clock, a Hamiltonian degree-of-freedom count with zero propagating scalar, and a healthy linear cosmology — and it explains why every viable construction in the programme is of cuscuton type. On the concrete cuscuton completion (an $F(Q)\Theta$ /rotated-constitutive clock, developed collaboratively in the public repository) we verify, with committed runnable scripts and a machine-checked Lean 4 kernel, that a single action reproduces: the galaxy radial-acceleration relation (SPARC, rms 0.16 dex at zero per-galaxy parameters), weak lensing tied to dynamics ($\Phi = \Psi$, no slip), cluster and cosmic-web dark matter as the conserved Noether charge of the clock’s shift symmetry (pressureless, clustering like cold dark matter with $\sigma_8 = 0.811$, and reproducing the Bullet-cluster lensing offset that pure MOND fails), a tensor gravitational-wave speed equal to c , and the Solar-System post-Newtonian bounds — including the preferred-frame parameter that excludes the earlier AeST completion, here structurally suppressed rather than un-tunable. We give the distinctive falsifiable predictions: a *flat* $a_0(z)$ (a $z \simeq 2.5$ baryonic-Tully–Fisher zero-point of 0.00 dex vs. +0.33 dex for Λ CDM), an External Field Effect in dwarf-spheroidal dispersions ($\sim 1.9\times$ across Galactocentric radius, where dark matter predicts none), a subdominant near-luminal scalar gravitational-wave polarization, a dark matter that cannot decay or annihilate, and a parameter-free $1/r$ correction above the flat rotation velocity. **We do not claim the theory is established.** Two costs are proved and stated: the charge-sourced dark matter carries an *intrinsic* ~ 24 -order Big-Bang-Nucleosynthesis fine-tuning (the same conserved charge sources an a^{-6} stiff partner, and the ghost-free degeneracy forbids removing it), and the full relativistic constraint algebra of the completion is still being derived. This is the honest state of a candidate: a consistent structural class, a phenomenology-complete instance, sharp predictions that data

can confirm or kill, and named open costs. Every dimensional number is carried on both a_0 footings. AI-assisted research draft; not peer reviewed.

1 The programme and the claim

Milgrom’s law fits galaxy rotation curves with one acceleration scale $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$ and no dark matter [1], but a relativistic completion that also fits lensing, clusters and the microwave background has been hard to build [2, 3]. This programme takes a_0 to be the de Sitter scale, $a_0 = c^2/2\pi L_{\text{dS}}$ with $L_{\text{dS}} = \sqrt{3/\Lambda}$ (verified to $\sim 5\%$ against the observed Λ), and asks what relativistic structure is forced. We report a classification theorem, a phenomenology-complete cuscutoon instance, its falsifiable predictions, and its proven costs. The result is stated as the honest status of a *candidate*: nothing here uses data to assert the theory is true.

2 The structural spine: closure forces a cuscutoon

The programme’s earlier no-go results showed that a single-metric MOND theory with two propagating modes forces a preferred frame [8], and that a clock-sourced dark component obeys an “excess-spent-once” overshoot unless the transmission ordering is reversed. The unifying structural fact is the following.

Theorem 1 (closure forces a cuscutoon) *Let the MOND sector be a canonical pair with Hamiltonian $\mathcal{H} = \frac{1}{2}A(s)p^2 + W(s)$, $s = u'$, $W'(s) = \mu(s)s$, μ the MOND interpolating kernel. The smeared hypersurface-deformation bracket is $\{H[N], H[M]\} = (NM' - MN')pA[\frac{1}{2}A'p^2 + \mu s]$. Closure onto the spatial-diffeomorphism generator $\propto ps$ requires both $A\mu = 1$ (so $A = 1/\mu$) and $A' = 0$; but $A = 1/\mu$ gives $A' = -\mu'/\mu^2$, so $A' = 0 \Leftrightarrow \mu' = 0$. A MOND kernel is strictly monotone ($\mu' \neq 0$), so no canonically-kinetic MOND scalar closes the algebra. The unique escape is the absence of the kinetic term, i.e. a **cuscutoon** (non-propagating scalar), on which the obstruction never arises and, having no independent kinetic mode, the scalar is automatically ghost-free.*

Thus *consistency itself*, before any stability question, selects the cuscutoon branch. This is the common root of the programme’s health results: the deep-MOND gradient instability is evaded only by keeping MOND inside the clock; the completion’s Hamiltonian degree-of-freedom count returns zero propagating scalar at $k \neq 0$; and its linear cosmology is healthy because the MOND operator is cubic in perturbations and does not add a propagating gradient mode. “Keep MOND inside the clock” is not a construction choice; it is the unique consistent option. The bracket and its cubic obstruction are reproduced in exact symbolic arithmetic; the theorem’s conclusion is the closure algebra of that bracket, general in μ .

3 The cuscutoon completion and what it reproduces

On the concrete cuscutoon clock (action schematically $S = \int \sqrt{-g}[\frac{M^2}{2}R - \Lambda M^2 - K(Q) + F(Q)\Theta + M^2 a_0^2 G(|V|/a_0)] + S_m$, with $G(y) = y^2 + 2(1+y)e^{-y} - 2$ so $G'(y)/2y = 1 - e^{-y}$; developed collaboratively in the public repository), a single action reproduces the following, each checked by a committed script:

Front	Result
Galaxy rotation (RAR)	Fits 175 SPARC curves at zero per-galaxy parameters, rms 0.16 dex; MOND-consistent baryonic Tully–Fisher.
Lensing	Static branch has $\Phi = \Psi$ (no slip): lensing follows dynamics on the same relation.
Dark matter	The conserved Noether charge of the clock’s shift symmetry: pressureless ($c_s^2 = 0$), cannot decay.
Structure growth	Linear growth is Λ CDM-identical ($c_s^2 = 0$, $G_{\text{eff}} = G$); $\sigma_8 = 0.811$, no deficit.
Clusters (Bullet)	Collisionless charge-dust traces galaxies; lensing separates from gas (pure MOND fails).
Gravitational waves	$c_T = c$ exactly; a subdominant near-luminal scalar polarization within current bounds.
Solar System	$\gamma = \beta = 1$, $\alpha_3 = 0$; the AeST-killing α_1 is structurally suppressed, not un-tunable.

4 Falsifiable predictions

(1) **Flat** $a_0(z)$: $a_0 \propto \sqrt{\Lambda}$ is constant, so the deep-MOND baryonic Tully–Fisher zero-point at $z \simeq 2.5$ is 0.00 dex versus +0.33 dex for Λ CDM; a robustly rising a_0 falsifies it. (2) **External Field Effect**: dwarf-spheroidal dispersions rise $\sim 1.9\times$ with Galactocentric distance (Crater II reproduced), where dark matter predicts none — the sharpest discriminator. (3) **Bullet offset**: as above. (4) **GW**: $c_T = c$ plus a subdominant scalar mode. (5) **No indirect detection**: the Noether-charge dark matter cannot decay or annihilate. (6) $1/r$ **rotation correction**: $v^2 = \sqrt{G_b M a_0} + G_b M/4r + 7(G_b M)^{3/2}/96r^2\sqrt{a_0} + \mathcal{O}(r^{-3})$, parameter-free.

5 The proven costs, stated honestly

The completion is not established, and two costs are proved. First, the charge-sourced dark matter carries an **intrinsic BBN fine-tuning**: on a Friedmann background the eliminated density is quadratic in the conserved charge C , $\rho = B + 3M^2H^2 - (M^2/3f^2)(A + C/a^3)^2$, so the same charge that sources the a^{-3} dust also sources a stiff a^{-6} term; BBN bounds the stiff piece to $\Omega_{\text{stiff},0} \lesssim 4 \times 10^{-25}$, forcing $|C|/|A| \lesssim 3 \times 10^{-24}$, and the ghost-free degeneracy (which fixes K quadratic) forbids a linear-in-charge, stiff-free dust — the tuning is not removable within the class. Second, the full relativistic constraint algebra of the completion (metric sector included) is still being derived; the scalar block closes on the cuscuton branch, but the complete health verdict, the cosmic-microwave-background transfer function, and the exact post-Newtonian calibration remain open. These are named, not hidden.

6 Reproducibility

Every claim is produced by a committed, runnable script with checks that can fail, in the public repository under `fable_independent_2026/` (the structural theorem, the phenomenology, the predictions, and the fine-tuning), and the load-bearing mathematical identities are machine-checked in Lean 4 (`fable_independent_2026/lean_2026/Mondlean.lean`, twelve theorems, standard axioms only). All dimensional numbers are carried on both a_0 footings ($9.3619 \times 10^{-11} \text{ m s}^{-2}$ and

$1.1279 \times 10^{-10} \text{ m s}^{-2}$). This is an AI-assisted research programme with a collaborative construction; it is not peer reviewed, and no observation is used to assert the theory is true.

References

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